

Voice Commands Tutorial

Drive every primitive, transform, boolean, and export with natural speech. No syntax to memorise — the LLM understands you.

Voice is the fastest way to model in ForgeSlicer once you've used it for an hour. Most users hit a 3× productivity bump over menu-clicking within their first week.

1. How voice control works

ForgeSlicer captures audio with your browser's microphone, sends it to **OpenAI Whisper-1** for transcription, then routes the resulting text through **GPT-5.2** to extract a structured intent (action + parameters). The intent runs against the same scene actions the toolbar buttons use, so anything you can click you can say.

Privacy & data flow

- Audio is streamed to OpenAI only while the mic button shows **Listening**. Nothing is recorded otherwise.
- Transcripts are NOT stored server-side. The browser keeps the last 20 in `localStorage` so you can scroll through your session, and clears them on sign-out.
- GPT-5.2 receives the transcript plus a short scene summary (currently-selected object IDs, modifier types, last action) so it can resolve pronouns like *'move it,' 'flip that one.'*

Heads-up — Universal API Key: Voice uses the Emergent Universal Key by default. If your account runs out of balance you'll see an amber banner; either top up at [Profile](#) → [Universal Key](#) or fall back to keyboard / mouse.

2. The mic button — three states

State	Look	Behaviour
Idle	Grey mic icon	Click (or press V) to start listening.
Listening	Pulsing orange mic	Speak now. Pause ~2 s and your transcript appears at the top of the workspace.
Processing	Spinner over the mic	Whisper is transcribing and GPT is parsing. Usually <1.5 s.
Confirming	Transcript banner	Say 'Run' to execute, or speak again to replace.

The mic is push-to-talk by default — start listening, speak, pause to commit. Hands-free mode (always-listening) lives behind [Profile](#) → [Voice](#) → [Always on](#).

3. Phrasing principles

GPT understands synonyms and verbs in any tense, so feel free to speak naturally. The principles below are habits that consistently get accurate intent parsing.

Be explicit about units when they matter

- ✓ *'Move it up ten millimeters'* — explicit.
- ✗ *'Move it up ten'* — ambiguous; the parser assumes mm but you might mean degrees on rotate.

Use pronouns — they work

ForgeSlicer feeds the LLM the currently-selected object IDs, so *'rotate it'*, *'make this taller'*, and *'delete that one'* all resolve to the active selection. Multi-select works too: select two parts, then *'subtract these'* runs CSG.

Chain in one breath

GPT parses compound commands when they're joined by 'and' or 'then'. Example: *'add a cube, then move it up ten, then duplicate it twice.'* The parser splits this into three sequential intents and replays them in order.

4. Lexicon — what you can say

The full searchable lexicon is in the in-app Help dialog (? → Voice Commands). Below is a quick reference grouped by category.

Add objects

Say...	What happens
'Add a cube'	Default positive cube (20×20×20 mm).
'Add a negative cylinder 5 mm wide'	Negative cylinder, diameter ≈ 5 mm.
'Drop a hex prism'	Cylinder with 6 sides.
'Add a 10 by 20 by 5 cube'	Cube with explicit XYZ dims.

Move / Rotate / Scale

Say...	What happens
'Move it up 10'	+10 mm on Y.
'Slide forward 3'	+3 mm on Z (toward camera).
'Rotate 90 degrees on Z'	Rotates about Z by 90°.
'Tilt 15 on X'	Rotates 15° about X.
'Make it twice as tall'	Scales Y × 2.
'Resize to 30 by 30 by 5'	Sets exact dimensions.
'Position at 0 10 0'	Absolute coordinates.

Scene management

Say...	What happens
'Drop to bed'	Snaps bottom to Y=0.
'Delete it'	Removes current selection.
'Duplicate and mirror on X'	Copy & mirror so the new part lands flush.
'Group these'	Wraps multi-selection in an Assembly.
'Ungroup'	Breaks selection out of its group.
'Select all'	Selects every object.
'Undo / Redo'	Atomic history step.

Booleans & modifiers

Say...	What happens
'Subtract these two'	CSG A – B on the two selected.
'Union them'	CSG union.

<i>'Intersect'</i>	CSG intersection.
<i>'Make this negative'</i>	Flip modifier tag.

5. AI generation by voice

Voice can fire the Meshy AI text-to-3D pipeline directly. There are two flavours — auto-submit (uses a credit immediately) and pre-fill (opens the dialog for review).

Trigger phrase	Behaviour
'Generate <noun> with AI'	Opens AI dialog, fills prompt, auto-submits — uses one credit.
'AI a <noun>'	Same as above, terse form. Auto-submits.
'Make me a <noun> with AI'	Auto-submits.
'I want to make a <noun> with AI'	Pre-fills, does NOT auto-submit — click Generate yourself to spend the credit.
'Open the AI generator'	Opens an empty dialog — no prompt, no credit used.

Auto-submit triggers cost one of your monthly credits (free tier: 13/month). The pre-fill phrasing is your safety net — use it when you're not sure the prompt will produce what you want.

6. Walkthrough — a phone stand by voice only

Goal: build a printable phone stand with a chamfered base and a negative groove for the phone, using only your voice.

- 1 Press **V**. Say 'Add a 70 by 50 by 8 cube'. Pause. Say 'Run.' A flat base appears.
- 2 Say 'Chamfer the edges by 2 millimeters.' The base gets lead-in bevels.
- 3 Say 'Add a negative cube 70 by 8 by 30.' A tall, thin negative slab appears.
- 4 Say 'Rotate it 15 degrees on X, then move it back 8 mm, then up 5 mm.' Three transforms chain into one slot for the phone.
- 5 Say 'Group these.' The stand becomes a single assembly.
- 6 Say 'Export as STL.' The browser downloads scene.stl — ready to slice.

7. Troubleshooting

Symptom	Likely cause	Fix
Mic button does nothing on click	Browser blocked microphone permission.	Click the camera/mic icon in the address bar → Allow.
Transcript is garbled / wrong language	Whisper auto-detects language; ambient noise or accents trip it.	Speak slightly slower; move away from fans; English-only phrasing currently has the best accuracy.
'I don't understand' banner	GPT didn't extract a valid intent.	Rephrase using a verb from the lexicon. Avoid filler words ('um', 'kind of').
Pronoun didn't resolve	Nothing was selected before the command.	Select first (click in scene or outliner), then say the command.
Amber 'no credit' banner	Universal Key balance depleted.	Top up at Profile → Universal Key.

8. Quick reference

Shortcut: V — toggle mic on/off.

Cancel: Esc while listening discards the current transcript without executing.

Help dialog: ? → Voice Commands — full lexicon with a Try ■ button next to every phrase so you can fire commands by click while learning.

Source files: frontend/src/lib/voice/ — the Whisper client, GPT prompt template, and intent → action router all live here.

*Tutorial generated May 29, 2026 from ForgeSlicer source. Run the corresponding scripts/build*_tutorial.py to regenerate after future updates.*